4.3

PAEDIATRICS

Central Nervous System Infections

CHAPTER 4.3	CENTRAL NERVOUS SYSTEM INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Meningitis			
< 1 month: Group B Streptococcus *E coli* Listeria (Rare)	Benzylpenicillin 100,000 units/kg/dose IV q8H (<7 days of life) q6H (>7 days of life) + Cefotaxime 50 mg/kg/ dose (max 2 g) IV q12H (<7 days of life) q6H (>7 days of life) **If Listeria suspected:** Ampicillin 50 mg/kg/ dose IV		• Duration of treatment depends on organisms: ▪ Group B Streptococcus: 14 days ▪ *S pneumoniae:* 14 days ▪ *N meningitidis:* 7 days ▪ *H influenzae:* 10 days ▪ Listeria: 21 days • For *S pneumoniae* which has high MIC for Penicillin (>2 ug/mL in meningitis) **ADD** Vancomycin 20 mg/kg IV q8H (>28 days old) to Ceftriaxone.
1 - 3 months: Organism from either age group			

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	q12H (<7 days old) q8H (7 - 21 days old) q6H (>21 days old) + Gentamicin 5 mg/kg/ dose IV q24H (Consider stopping Gentamicin after 7 days)		
>3 months: *S pneumoniae* *H influenzae* *N meningitidis*	Ceftriaxone 50 mg/kg/ dose (max 2 g) IV q12H **OR** Cefotaxime 50-75 mg/kg/dose (max 2 g) IV q6H		
Brain Abscess			
Primary Streptococci Anaerobic Gram-Negative Bacilli Enterobacteriaceae	Ceftriaxone 50 mg/kg/ dose (max 2 g) IV q12H +	Cefotaxime 50 mg/kg/ dose (max 2 g) IV q12H (< 7 days of life); q6H (>7 days of life)	• *Use Ceftazidime if *P aeruginosa* is suspected.

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S aureus Mixed Flora	Metronidazole 15 mg/kg/dose IV single dose followed by 7.5 mg/kg/dose IV q8H	**OR** Ceftazidime 50 mg/kg/ dose IV q8H*	
Post-Surgical, Traumatic Brain Abscess Staphylococci Pseudomonas Gram-negatives *S pneumoniae*	Cloxacillin 50 mg/kg/ dose IV q4-6H + Ceftriaxone 50 mg/kg/ dose IV q12H + Metronidazole 15 mg/kg/dose IV single dose followed by 7.5 mg/kg/dose IV q8H	Vancomycin 15 mg/kg/ dose IV q6-8H* Ceftazidime 50 mg/kg/ dose IV q8H**	• *Use Vancomycin if MRSA/MRSE is suspected. • **Use Ceftazidime if *P aeruginosa* is suspected.
Encephalitis			
Herpes simplex	**>3 months old:** Aciclovir 10 mg/kg/dose IV q8H **Neonatal herpes:**		• Duration: For at least 21 days in encephalitis - confirm cerebrospinal fluid negative for herpes simplex virus before stopping treatment.

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	Aciclovir 20 mg/kg/dose IV q8H		
Ventriculitis / Meningitis Following Ventriculoperitoneal Shunts			
S epidermidis *S aureus* Gram-negative *Acinetobacter baumannii*	Vancomycin 15 mg/kg/dose IV q6-8H + Ceftazidime 50 mg/kg/dose IV q8H	Vancomycin 15 mg/kg/dose IV q6-8H + Cefepime 50 mg/kg/dose q8H **OR** Meropenem 40 mg/kg/dose IV q8H	• For MSSA, deescalate Vancomycin to Cloxacillin 50 mg/kg/dose IV q4-6H. • In some situations, direct intraventricular instillation of targeted antimicrobial is needed in ventriculitis. Consult ID.